

Acids and Bases

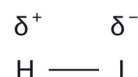
Name _____ Date _____

Use these principles to identify acids and bases, compare the acidity of organic compounds, and predict the favored direction of proton transfer reactions.

Acid and base definitions

Brønsted–Lowry acid: Donates a proton (H^+). Most of the time we won't see actual H^+ , it will be hydrogen attach to an electronegative atom.

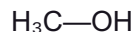
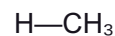
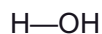
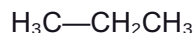
Examples:



In all cases, presence of more electronegative atom means the hydrogen is having its electrons “pulled away”. You can imagine them as vacuuming electrons away from the H, making it behave like H^+ .

Try it with bond polarity

Classify each displayed bond as ionic, nonpolar covalent, or polar covalent. For each polar covalent bond, place δ^+ and δ^- above the atoms and draw a crossed-tail dipole arrow.



After marking the dipoles, circle the bonds in which H is δ^+ . Those are the hydrogens to examine first as possible proton donors.

Brønsted–Lowry base: Accepts a proton (H^+).

Bases are the counterpart of the acid. When the hydrogens are experiencing electrons being pulled away, they're "drawn" to other things that could provide it with some electrons. Bases are typically described to have lots and lots of electrons around it.

Examples: NaOH, KOEt. It is recommended to draw out the charge separated compounds to see the actual base.

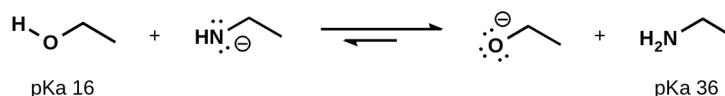
Hints for identifying the bases: will usually be paired up with metal ions. For the most part any time you see $\text{Na}^+ \text{K}^+$, you are hinted that its “partner” will be negatively charged thus basic.

Comparing the acidity of organic compounds

A lower pK_a indicates a stronger acid. An acid is generally stronger when the conjugate base formed by losing H^+ is more stable.

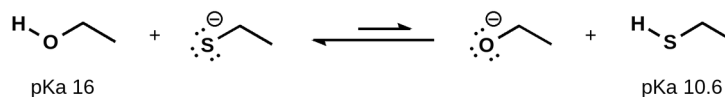
Consider the following factors when comparing conjugate bases. Evaluate them together, because more than one factor may affect stability.

Electronegativity: For atoms in the same period (row), a negative charge is generally more stable on the more electronegative atom.



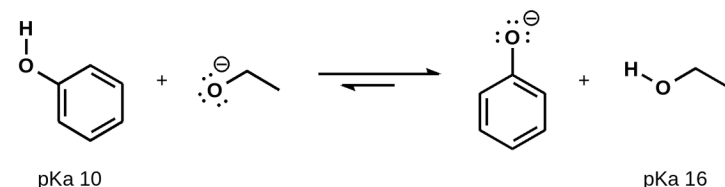
Example: O is more electronegative than N, so ethoxide holds the negative charge better than the amide ion. Ethanol (pK_a 16) is the stronger acid, and the equilibrium favors the right.

Atomic size: For atoms in the same group (column), a negative charge is generally more stable on the larger atom.



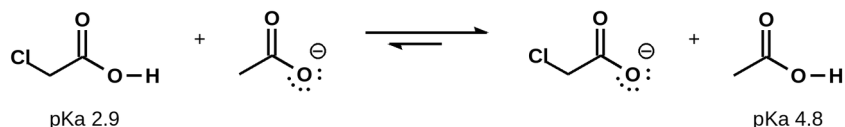
Example: S is larger than O, so the negative charge on S is spread over a bigger volume. The thiol (pK_a 10.6) is the stronger acid, so this equilibrium favors the left.

Resonance: A conjugate base is stabilized when its negative charge can be spread over multiple atoms through resonance.



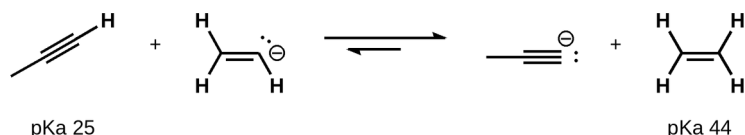
Example: Phenoxide spreads its negative charge into the ring by resonance; ethoxide cannot. Phenol (pK_a 10) is the stronger acid, and the equilibrium favors the right.

Inductive effects: Nearby electron withdrawing groups stabilize a negative charge by pulling electron density through bonds. This effect weakens with distance.



Example: The Cl pulls electron density through the bonds and stabilizes the carboxylate. Chloroacetic acid (pK_a 2.9) is stronger than acetic acid (pK_a 4.8), so the equilibrium favors the right.

Hybridization: For a negative charge on carbon, stability generally follows $sp > sp^2 > sp^3$. Greater s character holds the electron pair closer to the nucleus.



Example: The alkyne carbon is sp (50% s character), so its lone pair sits closer to the nucleus than in the sp^2 vinyl anion. Propyne (pK_a 25) is far more acidic than ethylene (pK_a 44), so the equilibrium favors the right.

Predicting acid base equilibrium

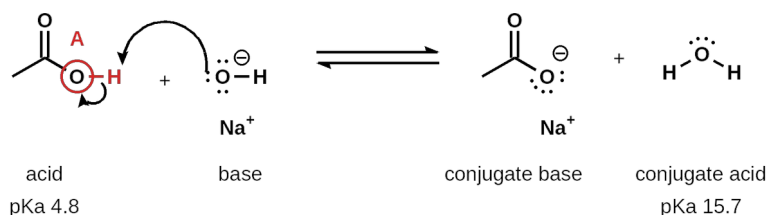
1. Write the complete proton transfer reaction and identify the acid, base, conjugate acid, and conjugate base.
2. Compare the two acids using their pK_a values, or compare the stability of the base and conjugate base.
3. Equilibrium favors the side containing the weaker acid (higher pK_a) and weaker, more stable base. Similar pK_a values indicate that neither side is strongly favored.

Worked example: acetic acid + NaOH

Every acid can be written as **H–A**: **H** is the proton that leaves, and **A** is the atom it is bonded to. When the H leaves, A keeps the two electrons from the H–A bond. In acetic acid, A is the O of the O–H.

How I draw the arrows for any acid–base reaction:

1. First identify the acid (the **H–A**) and the base (the atom with a lone pair or a negative charge).
2. Draw the first arrow from the lone pair on the base to the acidic **H**.
3. Draw the second arrow from the **H–A** bond onto **A**, the atom that keeps those electrons.
4. Label the conjugate acid and conjugate base, then compare pK_a values. Equilibrium favors the weaker acid.



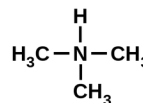
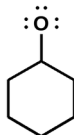
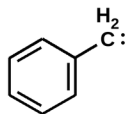
Acetic acid (pK_a 4.8) is a much stronger acid than water (pK_a 15.7), so this reaction lies far to the right. Na^+ is a spectator ion; it just comes along with the hydroxide.

Practice problems

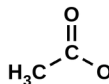
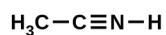
Work through these in order. Each set builds on the principles above. Show your reasoning, not just the answer.

1. Formal charge warm-up

Determine the formal charge on each bold atom below. Lone pairs have been drawn in for you.



Determine the formal charge on each bold atom below. This time you draw in the lone pairs.



2. Lewis structures of ions

Draw a Lewis structure for each ion. A heteroatom with a $\ominus$ charge has one fewer bond and one more lone pair than usual. A heteroatom with a $\oplus$ charge has one more bond and one fewer lone pair than usual.



(a) $\ominus$ on oxygen



$\oplus$ on nitrogen



$\oplus$ on carbon



(a) $\ominus$ on nitrogen

(b) $\ominus$ on carbon

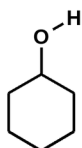
(b) $\ominus$ on carbon

3. Which is more acidic?

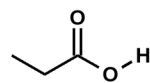
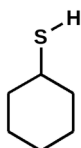
For each pair, circle the more acidic compound and write the factor that decided it (electronegativity, atomic size, resonance, induction, or hybridization).



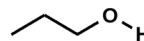
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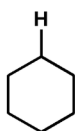
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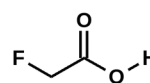
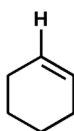
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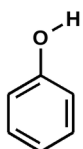
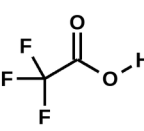
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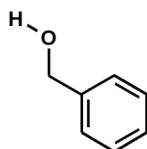
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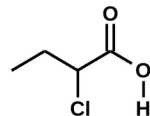
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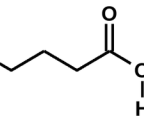
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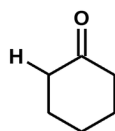
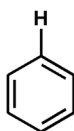
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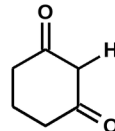
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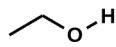
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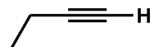
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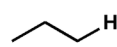
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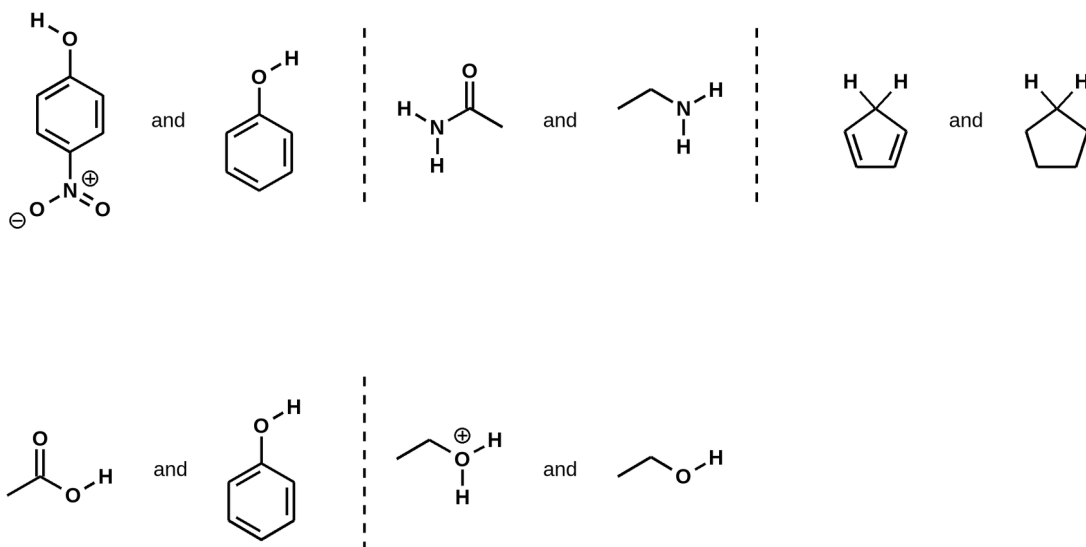
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A few more:



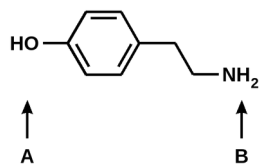
4. Ranking acidity

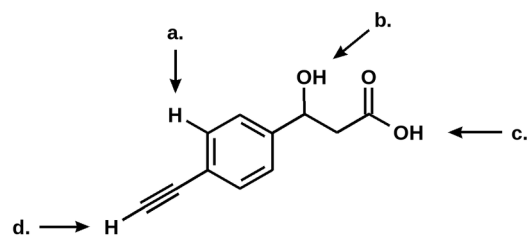
Rank each set in order of increasing acidity (weakest acid first).

- CH_4 HF NH_3 H_2O
- H_2Se H_2O H_2S
- NH_3 NH_4^+ NH_2^-
- $\text{CH}_3\text{CH}_2\text{OH}$ CH_3COOH $\text{CF}_3\text{CH}_2\text{OH}$
- $\text{HC}\equiv\text{CH}$ $\text{H}_2\text{C}=\text{CH}_2$ $\text{H}_3\text{C}-\text{CH}_3$

5. Most acidic hydrogen

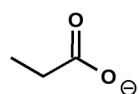
For each compound, decide which labeled hydrogen is the most acidic and estimate its pK_a .





6. Comparing base strength

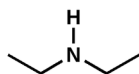
For each pair, circle the stronger base. Hint: first draw in all the lone pairs on each structure so you can see where the electrons are. Then compare: the stronger base has the weaker conjugate acid.



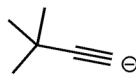
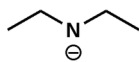
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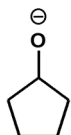
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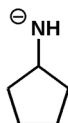
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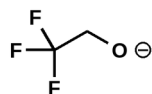
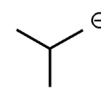
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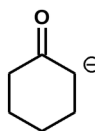
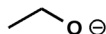
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or



or



or



Which species below is the strongest base?



a



b



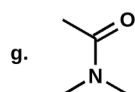
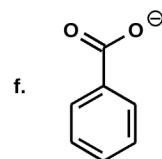
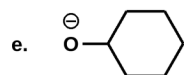
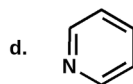
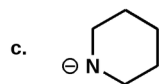
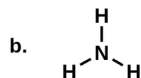
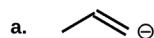
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d

7. Conjugate acids

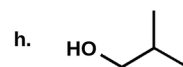
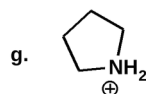
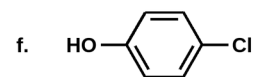
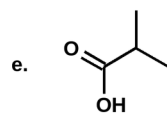
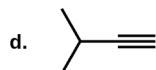
Draw the conjugate acid for each base.



In g there are two possible sites for protonation. Which one is preferred, and why?

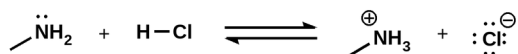
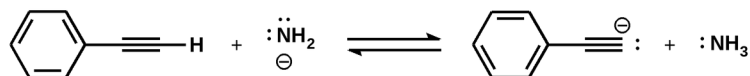
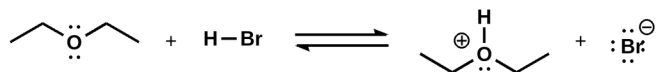
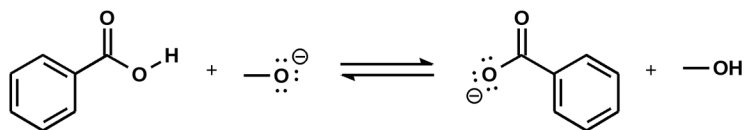
8. Conjugate bases

Draw the conjugate base for each compound.



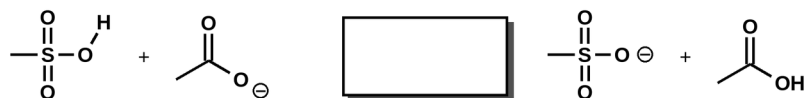
9. Curved arrows in acid–base reactions

Use curved arrows to show the flow of electrons in each reaction. Follow the same steps as the worked example: find the acid and base first, then draw arrows from the base to the acidic H.



10. Which side is favored?

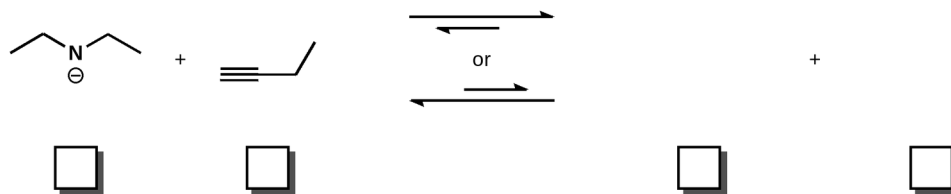
For each reaction, draw uneven equilibrium arrows in the box to show whether reactants or products are favored. Compare the two acids (or the two bases) to decide.



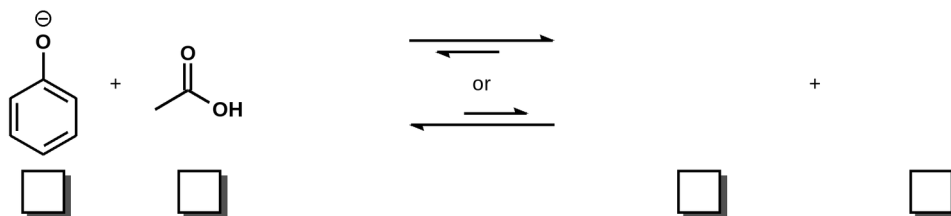
11. Complete the acid–base reaction

For each reaction: (1) draw the products, (2) label the acid (A), base (B), conjugate acid (CA), and conjugate base (CB) in the boxes, and (3) circle the set of equilibrium arrows that shows which side is favored.

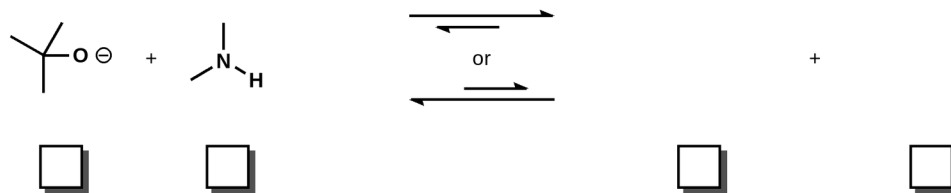
a.



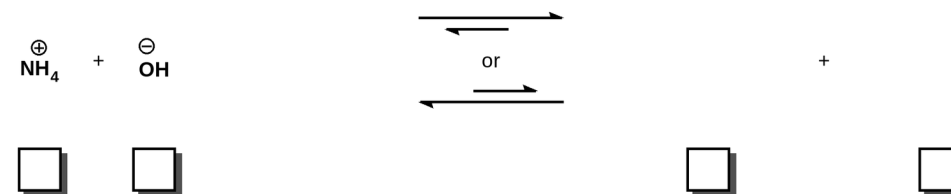
b.



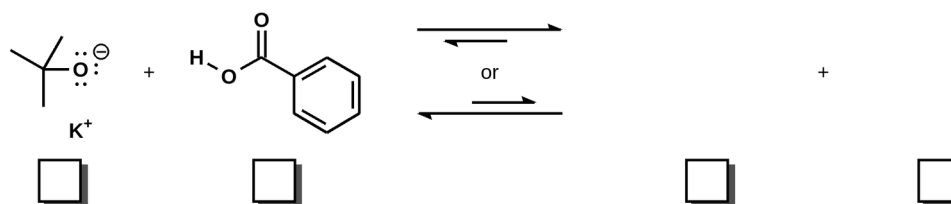
c.



d.



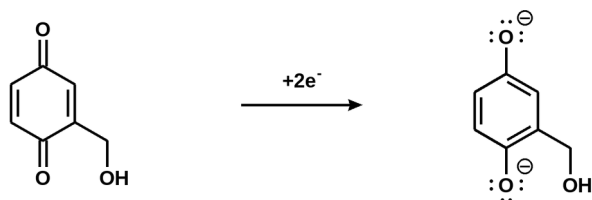
e.



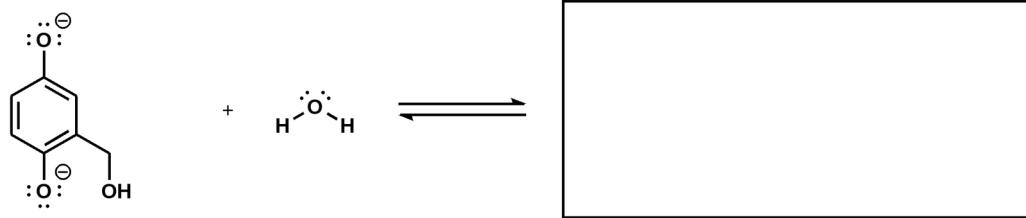
12. Acid–base chemistry in my research: quinones and CO₂ capture

In my PhD research I worked on molecules called quinones that can capture CO₂. When a quinone picks up two electrons it turns into a strong base, and that base is what grabs CO₂ or raises the pH of water. These problems use the actual molecules from that work, starting from 2-(hydroxymethyl)-1,4-benzoquinone, which I call HMQ.

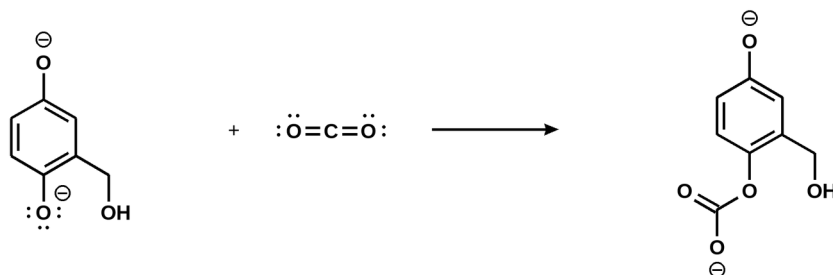
Reducing HMQ with two electrons gives the dianion shown. Circle the most basic atoms and explain why the dianion is a much stronger base than the neutral quinone.



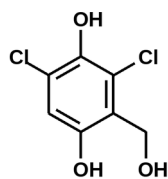
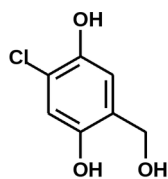
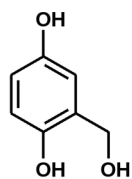
Complete the proton transfer between the reduced quinone and water. Using pK_a ≈ 11.4 for the hydroquinone monoanion (HQ⁻) and 15.7 for water, which side is favored? Why does the solution still become strongly basic?



Draw curved arrows for CO₂ capture by the reduced quinone. Label the Lewis acid and the Lewis base.



Rank these reduced quinones from least to most acidic (phenol O–H). Explain using the inductive effect. Would adding Cl make the reduced quinone a better or worse base for capturing CO₂?



Challenge: Reduced HMQ has two phenol O–H groups, labeled a and b. Which one is deprotonated first? Draw the conjugate base from each and explain your choice. (Hint: look at what sits next to each O–H.)

